SimpleDB, parte 6

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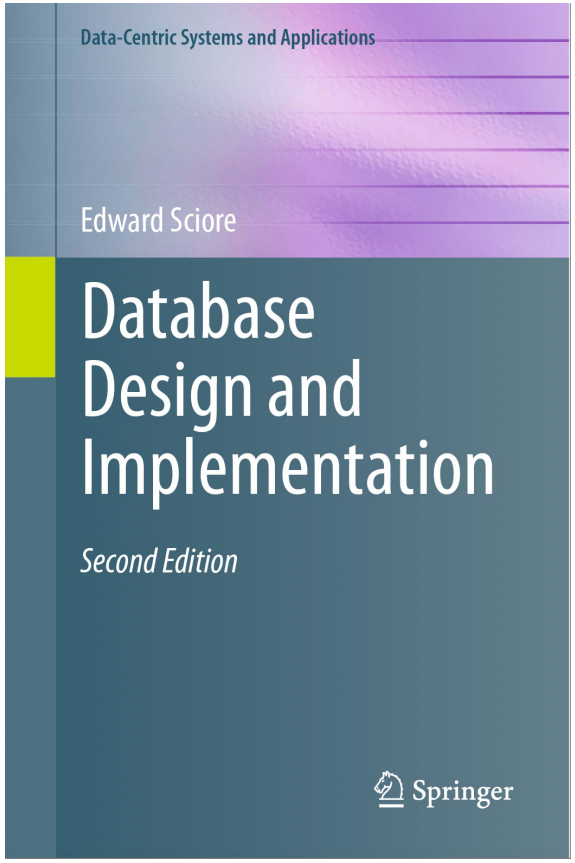


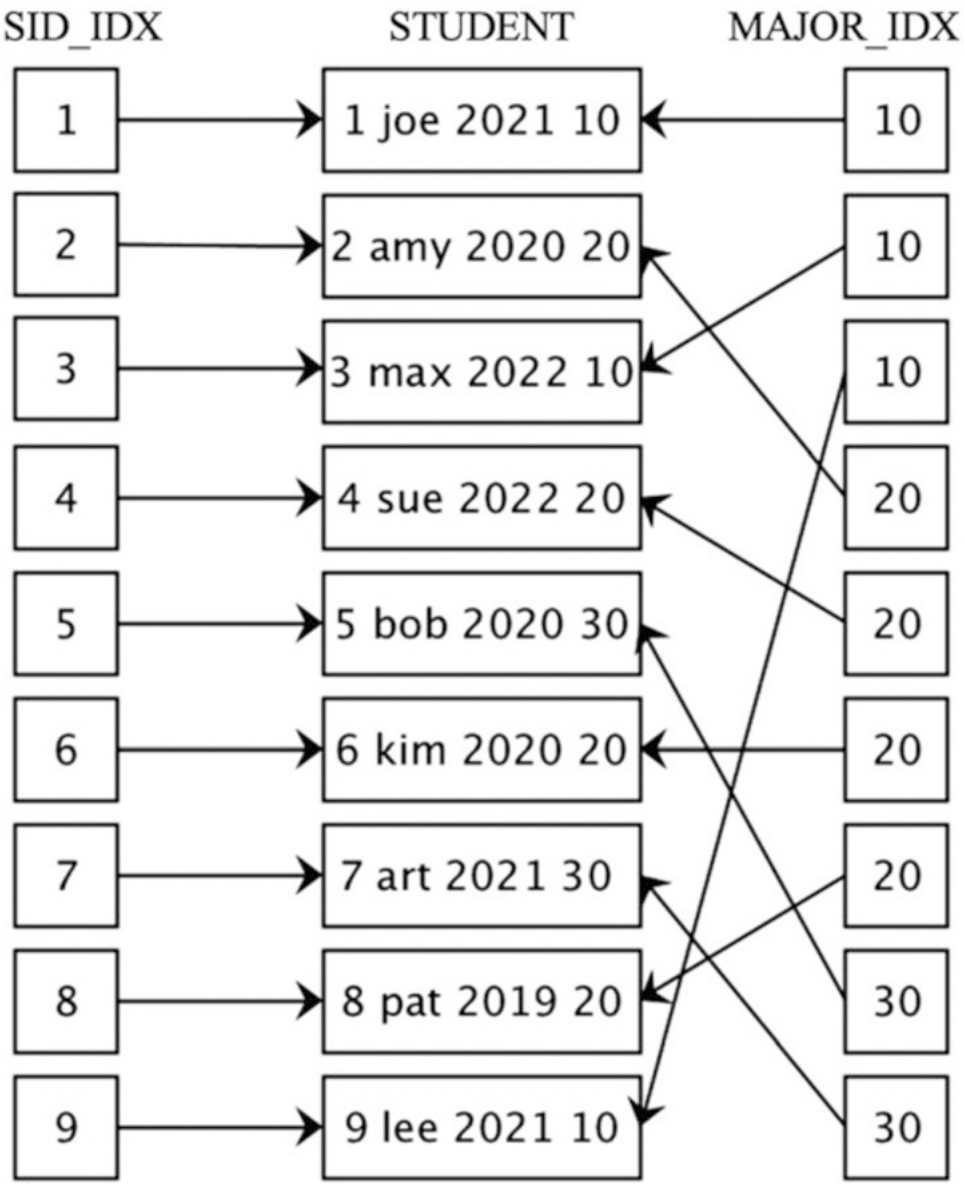
SimpleDB, package per migliorare le prestazioni

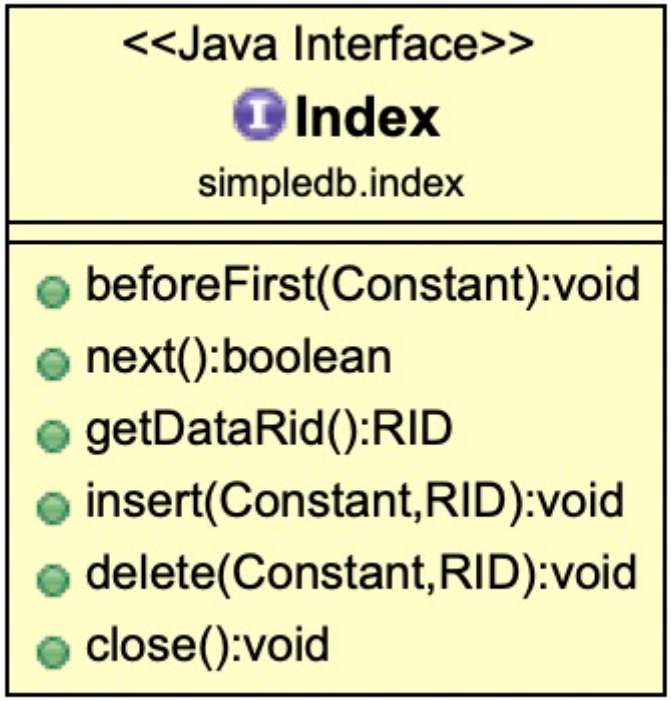
Index: Indici ed accesso diretto

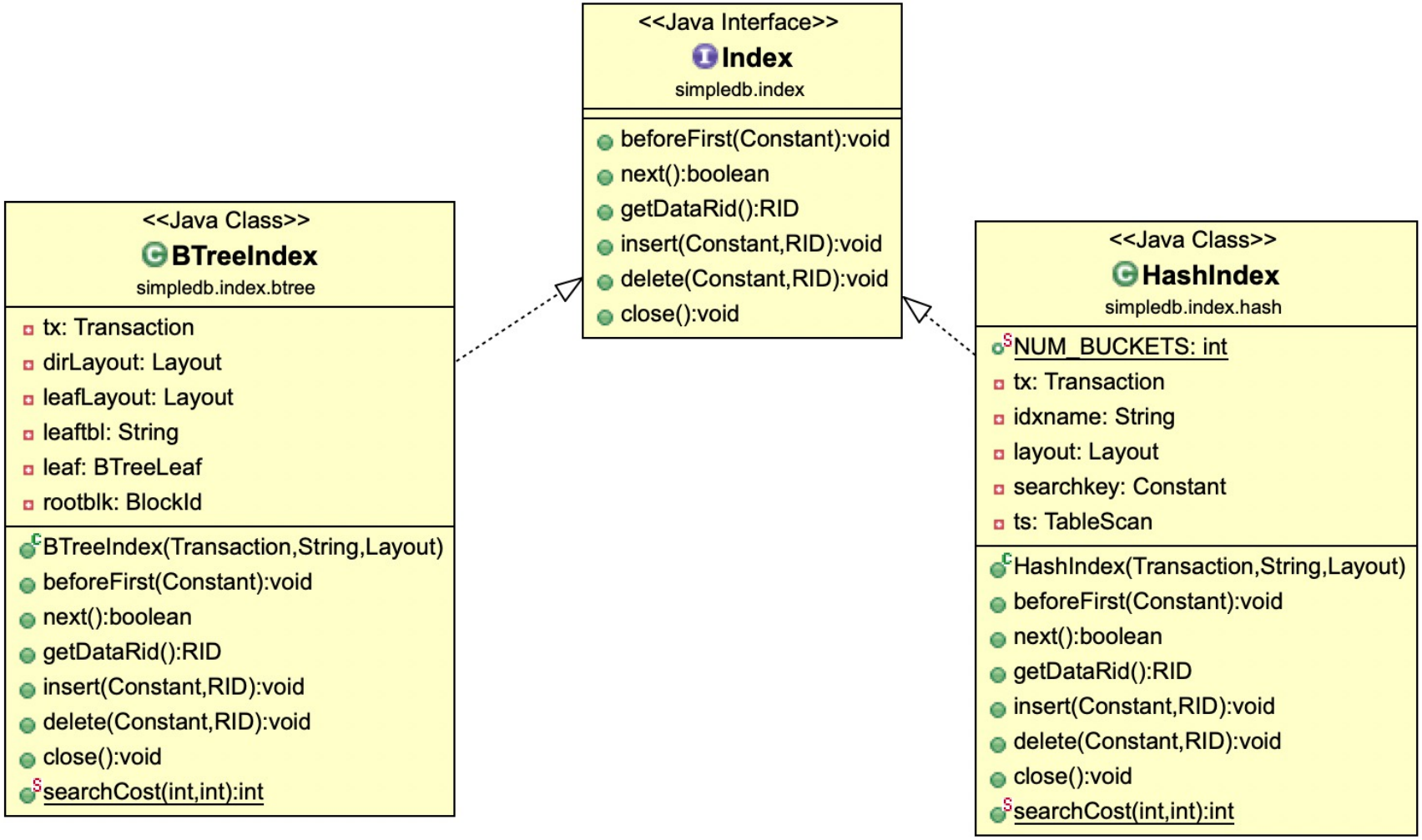
Materialize: Materializzazione e ordinamento

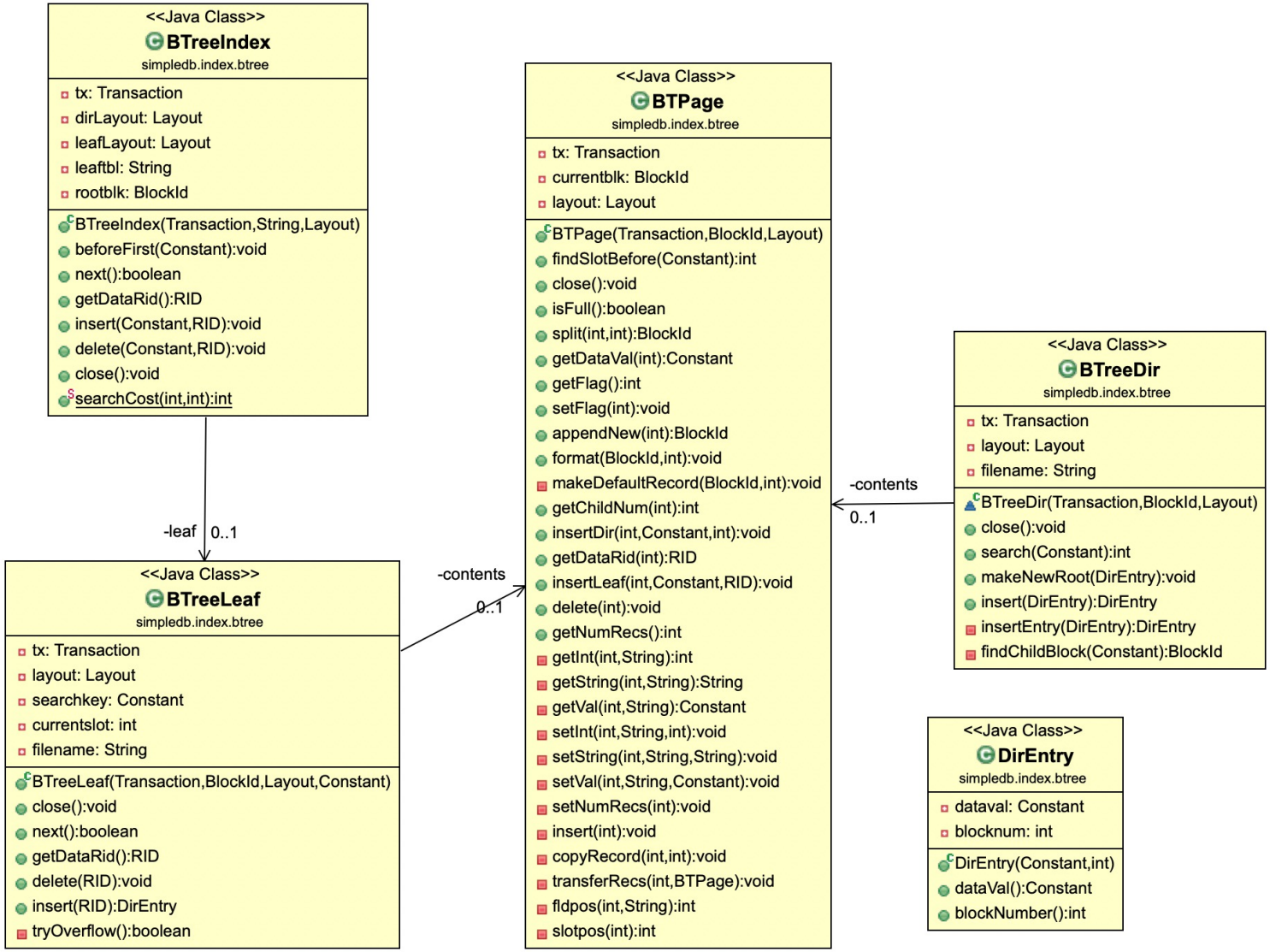
Multibuffer: Utilizzo efficace dei buffer

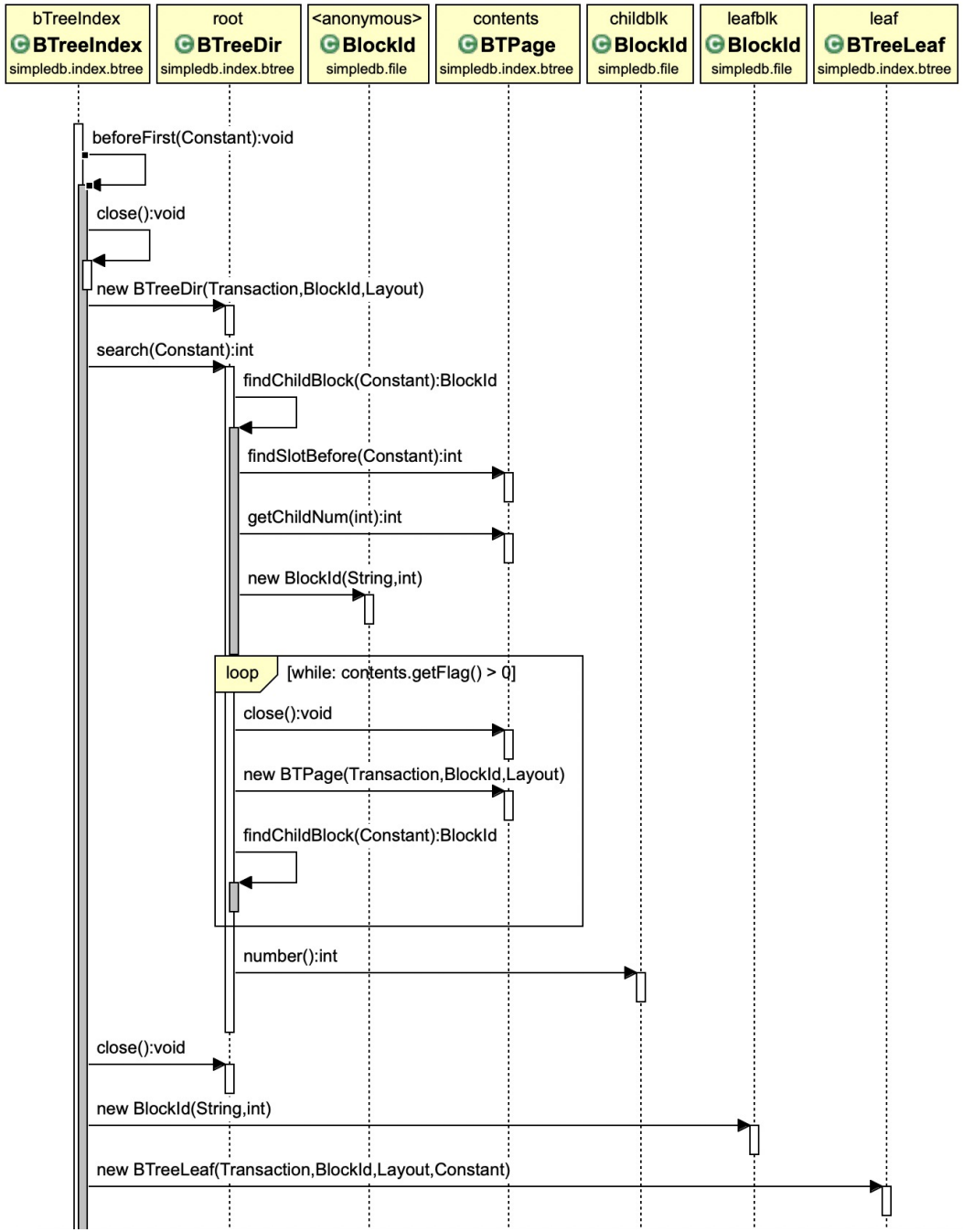
Opt: Ottimizzazione delle interrogazioni

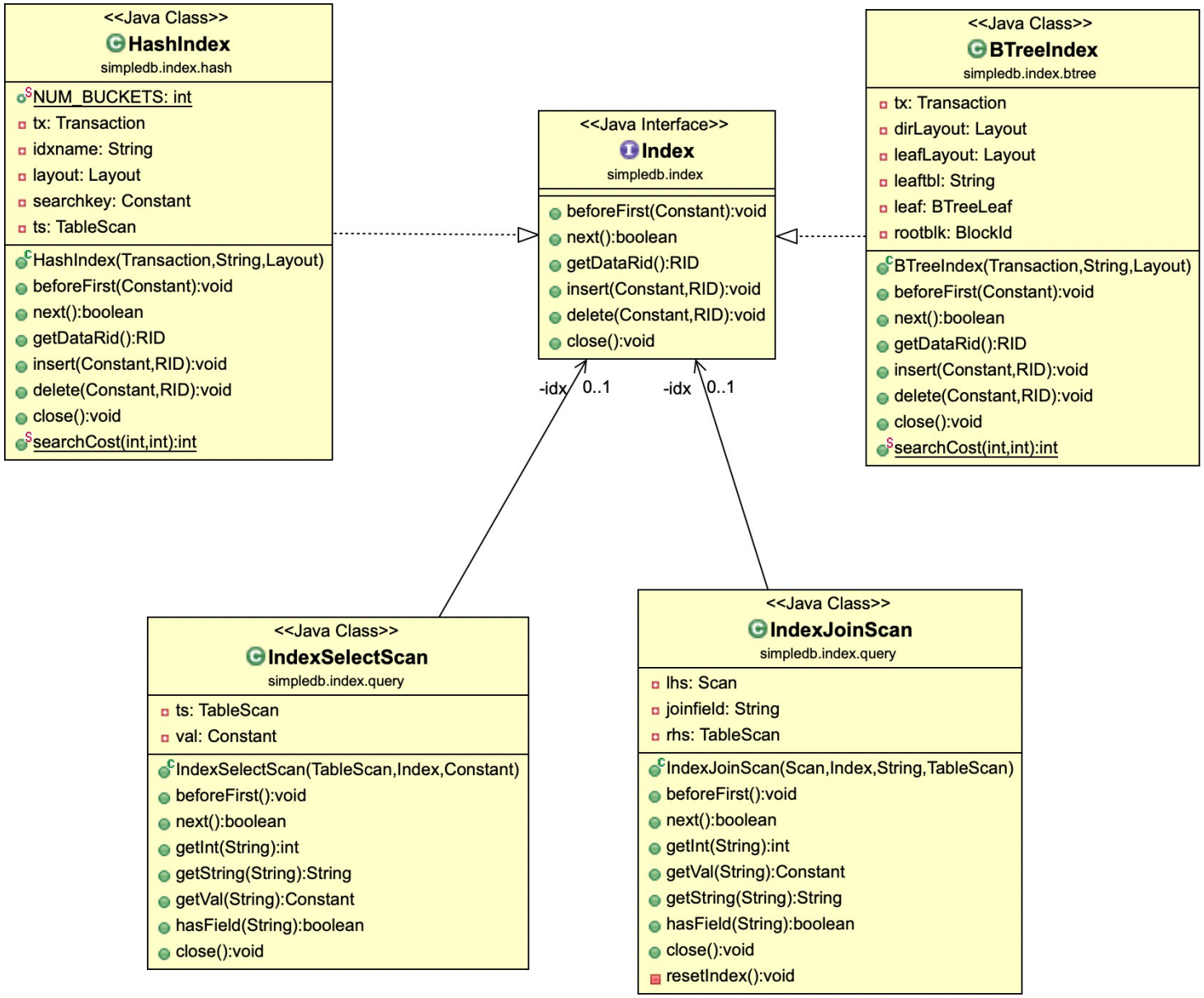

SID_IDX
STUDENT
MAJOR_IDX
1
2
3
4
5
6
7
8
9
1 joe 2021 10
2 amy 2020 20
3 max 2022 10
4 sue 2022 20
5 bob 2020 30
6 kim 2020 20
7 art 2021 30
8 pat 2019 20
9 lee 2021 10
10
10
10
20
20
20
20
30
30


<<Java Interface>>
Index
simpledb.index
beforeFirst(Constant):void
next():boolean
getDataRid():RID
insert(Constant,RID):void
delete(Constant,RID):void
close():void


<<Java Interface>>
Index
simpledb.index
beforeFirst(Constant):void
next():boolean
getDataRid():RID
insert(Constant,RID):void
delete(Constant,RID):void
close():void
<<Java Class>>
BTreeIndex
simpledb.index.btree
tx: Transaction
dirLayout: Layout
leafLayout: Layout
leaftbl: String
leaf: BTreeLeaf
rootblk: BlockId
BTreeIndex(Transaction,String,Layout)
beforeFirst(Constant):void
next():boolean
getDataRid():RID
insert(Constant,RID):void
delete(Constant,RID):void
close():void
searchCost(int,int):int
<<Java Class>>
HashIndex
simpledb.index.hash
NUM_BUCKETS: int
tx: Transaction
idxname: String
layout: Layout
searchkey: Constant
ts: TableScan
HashIndex(Transaction,String,Layout)
beforeFirst(Constant):void
next():boolean
getDataRid():RID
insert(Constant,RID):void
delete(Constant,RID):void
close():void
searchCost(int,int):int


<<Java Class>>
BTreeIndex
simpledb.index.btree
tx: Transaction
dirLayout: Layout
leafLayout: Layout
leaftbl: String
rootblk: BlockId
BTreeIndex(Transaction,String,Layout)
beforeFirst(Constant):void
next():boolean
getDataRid():RID
insert(Constant,RID):void
delete(Constant,RID):void
close():void
searchCost(int,int):int
-leaf
0..1
<<Java Class>>
BTreeLeaf
simpledb.index.btree
tx: Transaction
layout: Layout
searchkey: Constant
currentslot: int
filename: String
BTreeLeaf(Transaction,BlockId,Layout,Constant)
close():void
next():boolean
getDataRid():RID
delete(RID):void
insert(RID):DirEntry
tryOverflow():boolean
-contents
0..1
<<Java Class>>
BTPage
simpledb.index.btree
tx: Transaction
currentblk: BlockId
layout: Layout
BTPage(Transaction,BlockId,Layout)
findSlotBefore(Constant):int
close():void
isFull():boolean
split(int,int):BlockId
getDataVal(int):Constant
getFlag():int
setFlag(int):void
appendNew(int):BlockId
format(BlockId,int):void
makeDefaultRecord(BlockId,int):void
getChildNum(int):int
insertDir(int,Constant,int):void
getDataRid(int):RID
insertLeaf(int,Constant,RID):void
delete(int):void
getNumRecs():int
getInt(int,String):int
getString(int,String):String
getVal(int,String):Constant
setInt(int,String,int):void
setString(int,String,String):void
setVal(int,String,Constant):void
setNumRecs(int):void
insert(int):void
copyRecord(int,int):void
transferRecs(int,BTPage):void
fldpos(int,String):int
slotpos(int):int
-contents
0..1
<<Java Class>>
BTreeDir
simpledb.index.btree
tx: Transaction
layout: Layout
filename: String
BTreeDir(Transaction,BlockId,Layout)
close():void
search(Constant):int
makeNewRoot(DirEntry):void
insert(DirEntry):DirEntry
insertEntry(DirEntry):DirEntry
findChildBlock(Constant):BlockId
<<Java Class>>
DirEntry
simpledb.index.btree
dataval: Constant
blocknum: int
DirEntry(Constant,int)
dataVal():Constant
blockNumber():int


bTreeIndex
BTreeIndex
simpledb.index.btree
root
BTreeDir
simpledb.index.btree
<anonymous>
BlockId
simpledb.file
contents
BTPage
simpledb.index.btree
childblk
BlockId
simpledb.file
leafblk
BlockId
simpledb.file
leaf
BTreeLeaf
simpledb.index.btree
beforeFirst(Constant):void
close():void
new BTreeDir(Transaction,BlockId,Layout)
search(Constant):int
findChildBlock(Constant):BlockId
findSlotBefore(Constant):int
getChildNum(int):int
new BlockId(String,int)
loop
[while: contents.getFlag() > 0]
close():void
new BTPage(Transaction,BlockId,Layout)
findChildBlock(Constant):BlockId
number():int
close():void
new BlockId(String,int)
new BTreeLeaf(Transaction,BlockId,Layout,Constant)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NUM_BUCKETS: int
tx: Transaction
idxname: String
layout: Layout
searchkey: Constant
ts: TableScan
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next():boolean
getDataRid():RID
insert(Constant,RID):void
delete(Constant,RID):void
close():void
searchCost(int,int):int
<<Java Interface>>
Index
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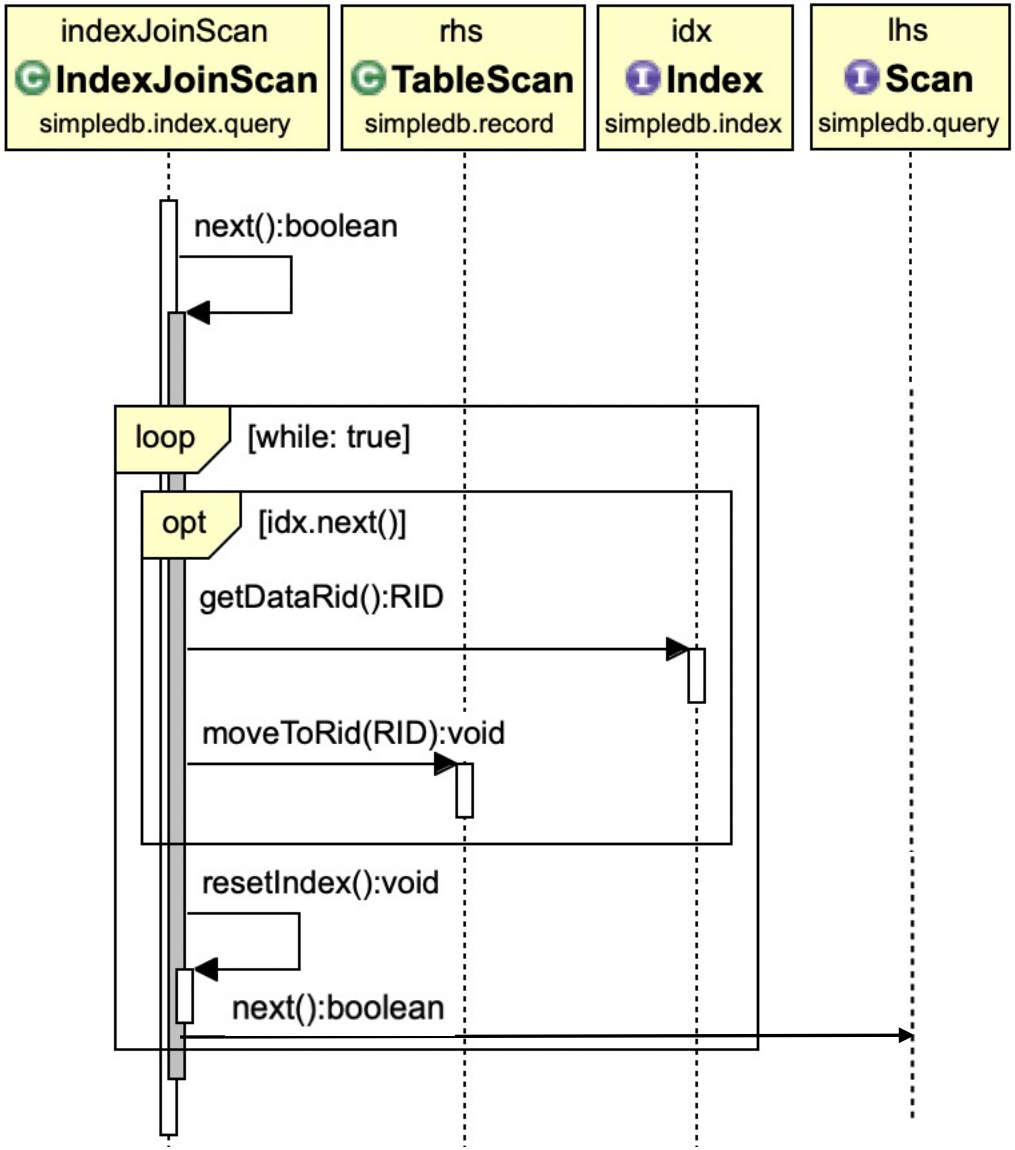
beforeFirst(Constant):void
next():boolean
getDataRid():RID
insert(Constant,RID):void
delete(Constant,RID):void
close():void
<<Java Class>>
BTreeIndex
simpledb.index.btree
tx: Transaction
dirLayout: Layout
leafLayout: Layout
leaftbl: String
leaf: BTreeLeaf
rootblk: BlockId
BTreeIndex(Transaction,String,Layout)
beforeFirst(Constant):void
next():boolean
getDataRid():RID
insert(Constant,RID):void
delete(Constant,RID):void
close():void
searchCost(int,int):int
-idx
0..1
-idx
0..1
<<Java Class>>
IndexSelectScan
simpledb.index.query
ts: TableScan
val: Constant
IndexSelectScan(TableScan,Index,Constant)
beforeFirst():void
next():boolean
getInt(String):int
getString(String):String
getVal(String):Constant
hasField(String):boolean
close():void
<<Java Class>>
IndexJoinScan
simpledb.index.query
lhs: Scan
joinfield: String
rhs: TableScan
IndexJoinScan(Scan,Index,String,TableScan)
beforeFirst():void
next():boolean
getInt(String):int
getVal(String):Constant
getString(String):String
hasField(String):boolean
close():void
resetIndex():void


indexJoinScan
IndexJoinScan
simpledb.index.query
rhs
TableScan
simpledb.record
idx
Index
simpledb.index
lhs
Scan
simpledb.query
next():boolean
loop
[while: true]
opt
[idx.next()]
getDataRid():RID
moveToRid(RID):void
resetIndex():void
next():boolean